Divya Upadhyay

divya.upadhyay0416@gmail.com | +1 (352) 721-1032 | **LinkedIn** | **GitHub** | **Portfolio**

Results-driven AI Software Engineer with 6 years of experience architecting and deploying scalable **Generative AI (GenAI/RAG)** systems and real-time **Computer Vision pipelines**. Specialist in productionizing **LLMs, MLOps**, and **AWS cloud** using automated CI/CD workflows and microservices architecture to optimize latency, throughput, and inference costs.

EXPERIENCE (6 years)

Website Developer | University of Florida, USA **| November 2025 – Present**

- Architected and launched the institution's enterprise website from scratch, establishing standardized **UF digital brand guidelines** and improving user engagement (UX) metrics across target audiences.
- Engineered modular **custom CSS and frontend style overrides** to extend native platform capabilities, ensuring 100% responsive design and visual compliance across all mobile and desktop viewports.
- Overhauled web accessibility framework to achieve full **WCAG 2.1 (Level A/AA)** compliance; conducted automated QA audits and technical **SEO optimizations** to boost organic search rankings.
- Directed enterprise **WordPress infrastructure administration**, managing server storage allocation, security patching, plugin ecosystems, and disaster recovery workflows that successfully recovered 5+ years of legacy visual assets.
- Integrated **Google Analytics** to evaluate user interaction patterns, driving data-informed decisions to expand digital audience reach and campaign performance.
- Designed brand-aligned digital marketing assets, speaker series graphics, campaign posters, and promotional logos using **Adobe Creative Cloud and Canva**.

AI Software Engineer | Golok Global INC, USA **| June 2025 – February 2026**

- Designed and deployed a real-time **conversational AI engine** for ride-hailing workflows, integrating **LLM orchestration and RAG-based vector search** to automate ride bookings and issue resolutions for thousands of daily active users.
- Engineered a high-throughput, low-latency payment processing microservice in **FastAPI with Redis caching** and idempotency keys, handling concurrent transaction traffic with sub-second internal processing prior to Stripe integration.
- Established an automated **MLOps platform** utilizing **GitHub, Jenkins, Docker, and AWS ECS**; cut model release latency by **95%** through automated CI/CD validation pipelines, artifact versioning, and zero-downtime rollback deployments.
- Spearheaded backend architecture for a travel platform by designing RESTful **API contracts, Java Spring Boot microservices, and AWS cloud hosting**; seamlessly integrated React frontend interfaces to support scalable user operations.

Data Scientist | Health Innovators INC, USA **| August 2024 – June 2025**

- Designed and deployed **LLM-powered** conversational pipelines for telehealth workflows using **OpenAI APIs** with structured **prompting**, **function calling**, and **RAG** over clinical FAQs, enabling automated symptom screening and patient triage; evaluated performance achieving an **18% increase in intent classification F1 scores and a 25% boost in resolution rates.**
- Architected a **multi-agent** clinical decision-support system using **LLM orchestration** (router + specialist agents, including a Radiology Agent) with report embedding, cross-agent context sharing, and chain-of-thought–constrained reasoning, improving clinician task completion time by **22%** and reducing manual report lookup latency by **35%**.
- Built a clinical NLP information extraction pipeline using Hugging Face Transformers (**fine-tuned BioClinicalBERT / PubMedBERT**) with **NER** and entity normalization to map unstructured notes and pathology reports to structured **FHIR-aligned schemas**, improving extraction **F1-score to 0.86** and increasing chart processing throughput by **60%**.
- Optimized analytical **SQL pipelines** supporting **daily ingestion of ~50GB of patient interaction data**, refactoring joins and indexing strategies to reduce query latency by **70%** and meet healthcare reporting **SLAs.**

Software Engineer (DevOps) | Nissan Digital India LLP, India **| September 2018 – June 2022**

- Engineered a Next-Gen delivery platform by building end-to-end **CI/CD pipelines with Jenkins, SonarQube, Veracode, and Nexus**; reduced software release cycle times by **40%** through automated quality gates and security scans.
- Automated multi-environment cloud infrastructure provisioning using **Ansible and AWS CloudFormation**, enabling zero-touch spin-up of dev, staging, and production environments with immutable infrastructure patterns.
- Implemented observability and alerting across critical services using **Prometheus, Grafana, and the ELK stack**, establishing SLO-driven monitoring and proactive alerting that reduced incident response time and improved platform availability.

SKILLS

AI & Generative AI: Generative AI, Large Language Models (LLMs), RAG, Multi-Agent Systems, Agentic Orchestration (LangGraph, AutoGen), PEFT/QLoRA, Model Fine-Tuning, Computer Vision, OpenCV, NLP, Transformers, PyTorch, TensorFlow, Hugging Face
MLOps & Cloud Infrastructure: MLOps, AWS (SageMaker, ECS, Lambda, EC2, CloudFormation), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), LangSmith, MLflow, Prometheus, Grafana, Infrastructure as Code (Ansible, Terraform)
Software & Data Engineering: Python, Java (Spring Boot), C++, SQL, FastAPI, RESTful APIs, Microservices, Vector Databases (Pinecone, ChromaDB, Qdrant), Redis, PySpark, Airflow, Git, Linux, WordPress

EDUCATION

- **MS in Computer and Information Sciences |** University of Florida **[May 2024] | GPA: 3.90 / 4.00**
- **B-Tech in Computer Science and Engineering |** KIIT University, India **[May 2019] | GPA: 8.98 / 10.00**